

This project investigates how different product characteristics influence user satisfaction in skincare products, focusing on the question: how do price and ethical/formulation features (such as clean, vegan, and cruelty-free labels) affect user ratings? The main hypothesis is that a product's overall ethical and ingredient safety status has a stronger effect on user satisfaction than its retail price, while the null hypothesis assumes no significant relationship between these variables and user ratings. The project will use multiple real-world datasets, including the following sources:

- Sephora products and reviews dataset (Kaggle)
- Cruelty-Free brands and retailers dataset (Kaggle)

Data will be collected by downloading these datasets from Kaggle and processed through several steps. This includes merging product, review, and brand ethics data using common identifiers (product IDs and brand names), and generating a new engineered feature called "safety score" based on whether a product is clean, vegan, or cruelty-free.

To correctly establish the relative importance of these factors using multiple linear regression, the independent variables will be standardized using StandardScaler in scikit-learn. This step is crucial because it allows for the direct and fair comparison of coefficients for variables measured on entirely different scales (e.g., price in dollars versus categorical safety scores). **The final dataset is expected to include approximately 2,350 skincare products and over 1 million user reviews**, with numerical variables (standardized price, rating, safety score). This structured dataset will enable the use of statistical analysis and regression models to evaluate the relative impact of each factor

and rigorously test the proposed hypothesis.